

CHEROKEE, IA, USA

WMO: 720344

Lat: 42.732N

Lon: 95.556W

Elev: 1226

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-3.0	-15.2	2.5	-6.7	-10.5	3.3	-2.4	25.5	23.9	23.7	24.4	3.2	0	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	92.0	73.8	89.5	72.9	86.4	71.4	78.3	86.3	76.5	84.5	74.9	82.8	11.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	143.7	83.3	73.8	132.0	81.2	72.5	125.9	79.7	42.8	85.5	40.8	84.9	39.3	82.6	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	20.2	18.2		-18.2	96.4	6.5	3.7	-22.9	99.0	-26.7	101.2	-30.3	103.2	-35.1	105.9
			DB	-19.4	81.7	6.1	1.2	-23.7	82.5	-27.3	83.2	-30.7	83.8	-35.1	84.7
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	21.4	21.5	37.3	48.0	60.0	70.9	74.5	71.2	64.0	50.2	37.0	24.3
	DBStd	21.46	13.01	12.12	12.96	10.31	9.23	6.24	6.15	5.64	8.00	9.46	10.37	11.25
	HDD50	3608	888	798	431	156	20	0	0	0	3	113	403	796
	HDD65	6876	1353	1217	861	516	209	20	5	12	114	468	840	1261
	CDD50	3063	0	0	37	96	331	626	759	659	423	120	12	0
	CDD65	857	0	0	2	6	55	196	299	206	83	10	0	0
	CDH74	8447	0	0	26	121	632	1946	3080	1769	760	110	3	0
	CDH80	2941	0	0	4	38	208	665	1260	539	202	25	0	0
Wind	WSAvg	7.5	8.0	7.9	8.8	9.7	8.4	6.9	5.8	5.3	6.3	7.6	7.9	7.6
Precipitation	PrecAvg	30.2	0.8	0.8	1.8	3.1	4.3	5.1	3.6	3.9	3.2	1.8	1.5	0.8
	PrecMax	36.4	1.5	1.9	4.0	5.7	7.1	8.6	7.6	10.8	7.7	4.6	4.3	1.7
	PrecMin	22.8	0.2	0.2	0.0	0.8	1.4	1.0	0.7	0.2	1.0	0.4	0.1	0.0
	PrecStd	4.1	0.4	0.4	1.2	1.3	1.5	2.1	1.5	2.8	1.6	1.2	1.1	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	58.5	77.9	86.6	91.3	93.6	98.2	93.7	90.0	83.9	70.5	54.3
		MCWB	42.8	49.4	58.0	61.2	67.3	73.4	74.5	74.6	70.4	66.8	56.0	45.0
	2%	DB	45.0	47.5	69.8	76.6	85.9	89.8	93.4	89.2	85.6	76.9	63.5	48.6
		MCWB	39.0	42.0	55.7	57.9	65.3	70.7	74.6	74.9	71.3	61.2	51.5	43.5
	5%	DB	40.6	42.4	62.8	70.9	81.3	87.1	90.3	85.8	81.8	71.6	57.8	43.1
		MCWB	36.5	37.9	54.5	55.7	64.0	69.5	75.2	73.8	68.2	59.7	48.2	38.3
	10%	DB	37.0	37.2	55.8	65.8	76.7	83.9	87.2	82.2	78.6	66.3	52.5	38.5
		MCWB	34.4	34.4	49.6	52.4	62.4	69.0	74.0	71.4	66.9	56.2	45.4	36.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	50.9	63.0	63.9	73.0	78.4	81.9	79.6	76.7	70.5	59.2	50.0
		MCDB	50.8	58.5	73.0	77.4	83.3	86.7	89.7	88.4	83.5	78.6	65.7	51.8
	2%	WB	39.4	42.2	59.1	60.5	70.0	75.6	79.3	77.4	74.5	65.0	53.8	42.3
		MCDB	43.6	47.7	64.6	71.7	79.7	83.8	87.1	85.1	80.8	73.5	62.7	46.8
	5%	WB	37.0	38.2	54.4	57.6	67.2	73.9	77.3	75.6	72.6	61.0	49.6	38.8
		MCDB	40.0	42.0	62.0	67.5	76.6	82.9	85.7	82.7	78.4	67.6	56.2	42.5
	10%	WB	34.8	34.9	49.4	54.7	64.6	71.7	75.4	73.8	70.3	57.1	45.9	36.3
		MCDB	36.9	37.4	57.0	64.1	72.2	80.3	84.1	80.2	75.4	64.0	52.5	39.0
Mean Daily Temperature Range	5% DB	MDBR	17.1	18.1	19.1	21.7	21.9	21.1	20.7	19.9	22.4	20.9	20.2	16.1
		MCDBR	21.2	23.5	29.7	30.9	27.7	24.2	23.0	22.3	24.6	27.7	28.2	21.9
		MCWBR	16.0	17.5	17.0	16.2	11.9	10.1	9.5	10.5	11.2	14.3	16.8	16.2
	5% WB	MCDBR	19.1	21.9	25.3	25.1	21.6	19.6	19.3	18.4	19.4	21.0	26.2	19.5
MCWBR		14.9	16.6	15.1	14.6	11.6	10.6	10.1	10.4	10.9	13.4	17.0	15.2	
Clear-Sky Solar Irradiance	taub	0.282	0.292	0.324	0.369	0.394	0.406	0.417	0.410	0.369	0.335	0.296	0.281	
	taud	2.404	2.394	2.398	2.299	2.300	2.325	2.280	2.291	2.373	2.450	2.484	2.439	
	Ebn at Noon	274	290	292	285	279	275	271	269	272	270	267	264	
	Edh at Noon	24	29	33	40	41	40	41	40	34	28	23	22	
All-Sky Solar Radiation	RadAvg	606	944	1213	1525	1789	1988	2091	1749	1414	967	662	497	
	RadStd	56	73	105	133	129	155	105	123	90	119	42	38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page